

Jinghang Li

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Summary

Research scientist specializing in machine learning and deep learning for large-scale, multimodal biomedical data. Experienced in developing, validating, and deploying end-to-end ML pipelines that transform research concepts into practical clinical and scientific impact.

Skills

Programming & Systems: Python, Bash, MATLAB, R

Machine Learning: Supervised and unsupervised learning, deep learning, classification, feature extraction, inference optimization, model explainability

ML Frameworks & Libraries: PyTorch, TensorFlow, Numpy, Pandas, scikit-learn, opencv, MONAI, TorchIO

Compute & MLOps: Linux, HPC/Slurm, multi-GPU DDP, Docker, Singularity, Git

Domain Expertise: Medical imaging (3T/7T MRI), MRI reconstruction (GRAPPA)

Projects

Multi-Modal MRI Enhancement: Conditioned Flow Matching from Raw k-space

Built an end-to-end pipeline from raw k-space to enhanced images, curating raw acquisitions, simulating physically grounded artifacts, and training acquisition-conditioned flow-matching models for multi-modal enhancement.

- Curated and organized raw Siemens TWIX (.dat) k-space across sequences (MPRAGE/TSE/SWI/TOF); built a GRAPPA reconstruction toolkit (recon/inspect/seq-info CLIs) with automatic 2D/3D detection and multi-GPU support.
- Wrote custom, physically grounded k-space artifact simulation: GRAPPA undersampling (R factors), TV-regularized compressed sensing, turbo-factor, and anisotropic-resolution downsampling.
- Conditioned a MONAI flow-matching UNet via cross-attention on acquisition metadata (TR/TE/TI, flip angle, bandwidth, field strength, voxel size) for controllable, multi-modal enhancement across contrast, artifact level, and field strength.
- Trained 3D conditional flow-matching models (multi-contrast joint training with balanced sampling, multi-GPU DDP, fp16, torch.compile, autoregressive single-step synthesis); enhances both fully sampled raw and post-processed (bias-corrected, denoised) images.

Autoregressive Flow Matching for Single-Step MR Image Contrast Translation

Led a zero to one, clinically focused pipeline for single-step MRI contrast translation from training dataset curation to on-prem GPU deployment.

- Built a large scale paired 7T MRI dataset; scaled processing via Slurm job arrays on HPC.
- Trained conditional diffusion and flow-matching models using PyTorch/MONAI; utilized multi-GPU DDP training on HPC for faster iteration.
- Productionized inference for NIfTI/DICOM with Docker/Singularity.
- Submitted invention disclosure to the university innovation office; provisional patent filed.

GPU-Accelerated RF Shimming Optimization (60-channel Tx)

Physics-informed optimization for large-scale RF shim parameter search under scanner power constraints.

- Implemented RF shimming optimization in PyTorch to leverage GPUs and Slurm for large-scale optimizations.
- Optimized amplitudes/phases across 60 transmit channels over large FOV volumes ($\approx 320 \times 320 \times 400$)
- Implemented differentiable histogram objective function from literature to improve solution quality vs prior baselines.

niimetric — Image-Quality Metrics & MRI Artifact Simulation (open source)

Personal open-source CLI toolkit (PyPI: niimetric) for fast quality evaluation of enhanced MRI and physics-grounded artifact simulation.

- Built a pip-installable CLI to benchmark enhanced images against references with PSNR, SSIM, MAE, and LPIPS (slice-wise, brain-masked, CSV export).

- Added custom, physics-grounded MRI artifact simulation (coil-based noise, GRAPPA, motion, ghosting, bias field, blur) at configurable severity, building on TorchIO.
- Published to PyPI under the MIT license for reproducible, scriptable image-quality evaluation.

Education

PhD	University of Pittsburgh , Bioengineering – Pittsburgh, PA, USA	Aug 2021 – present
	Carnegie Mellon University , Visiting Student, Computer Science – Pittsburgh, PA, USA	2021 – 2023
BS	University of Pittsburgh , Bioengineering – Pittsburgh, PA, USA	2016 – 2021

Publications

MRecover: A Conditional Generative Model for Recovering Motion-Corrupted MR Images Using AI Generated Contrast	May 2026
<i>Jinghang Li</i> , Tales Santini, Courtney Clark, et al., Howard Aizenstein, Minjie Wu, Tamer Ibrahim arxiv.org/abs/2605.21669 (arxiv, npj Artificial Intelligence [under review])	
Bridging Neuroimaging and Neuropathology: A Comprehensive Workflow for Targeted Sampling of White Matter Lesions	Oct 2025
<i>Jinghang Li</i> , Nadim Farhat, Jacob Berardinelli, et al., Minjie Wu, Julia Kofler, Tamer Ibrahim https://doi.org/10.1111/jon.70094 (Neuroimaging)	
A full life cycle biological clock based on routine clinical data and its impact in health and diseases	Oct 2025
Kai Wang, ..., <i>Jinghang Li</i> , ..., Xiangmei Chen, Kang Zhang www.nature.com/articles/s41591-025-04006-w (Nature Medicine)	
RF shimming strategy for an open 60-channel RF transmit 7T MRI head coil for routine use on the single transmit mode	May 2025
Andrea N. Sajewski, ..., <i>Jinghang Li</i> , ..., Tamer S. Ibrahim 10.1002/mrm.30563 (Magnetic Resonance in Medicine)	
wmh_seg: Transformer based U-Net for Robust and Automatic White Matter Hyperintensity Segmentation across 1.5 T, 3T and 7T.	Feb 2024
<i>Jinghang Li</i> , Tales Santini, Yuanzhe Huang, et al., Tamer Ibrahim, Howard Aizenstein, Minjie Wu arxiv.org/abs/2402.12701 (arxiv)	

Awards

Student Venture Big Idea Competition — 3rd Place \$5,000, University of Pittsburgh (April 2026)

Department Best Teaching Assistant (Nominated), University of Pittsburgh (2023)